

risa: Basic functions

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The *risa* package

The **risa** package, which stands for RISK Assessment, was designed to automate the application of the Habitat Risk Assessment (HRA) model, which is part of the *InVEST* (Integrated Valuation of Ecosystem Services and Tradeoffs) toolkit developed by the Natural Capital Project (Arkema et al., 2014; Sharp et al., 2020).

There are a entire suite of functions that were projected to automate the steps to apply the *ByRA* (Bycatch Risk Assessment) toolkit (Wilhelm et al., 2022). *ByRA* uses HRA methodology to quantify the spatial overlap between fisheries and marine megafauna and has been applied to assess bycatch risk for marine mammals.

Specifically, **risa** automates the generation of Kernel Density Maps of species occurrences and stressor intensities from either data frames or shapefiles, the creation of overlap hotspot maps, and the execution of the full HRA model for one or multiple species and stressors. Users can select among multiplicative and Euclidean formulations of risk, apply optional decay functions to account for stressor influence buffers, and generate detailed summary statistics. In addition, **risa** provides flexibility to customize model settings, including the number of risk classes, buffer sizes for kernel density estimates and stressor zones, and the choice of coordinate system (metric or decimal).

Installation

Since **risa** is still in its *alpha* version, it cannot be installed from R's CRAN repository. Thus, **risa** must be installed from a GitHub repository, which can be assessed via the link:

<https://github.com/GusmaoJB/risa/>

The easiest way to install **risa** is via `devtools::install_github()`:

```
# install devtools
library(devtools)

# install the risa package
devtools::install_github("GusmaoJB/risa")
```

Since we depend on function in the packages *terra* and *sf* to load and manipulate rasters and shapefiles, we need to load these packages to perform our examples. Also, let's keep our work files (rasters, shapefiles, dataframes) in a *data* folder inside our working directory. In this first example, let's suppose we have species and stressor distributions as occurrence points in CSV files:

```
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.10.3, PROJ 9.6.2; sf_use_s2() is TRUE
```

```
library(terra)
```

```
## terra 1.8.93
```

```
#spp_df <- read.csv("data\species.csv")
#str_df <- read.csv("data\stressor.csv")

#head(spp_df)
#head(str_df)
```

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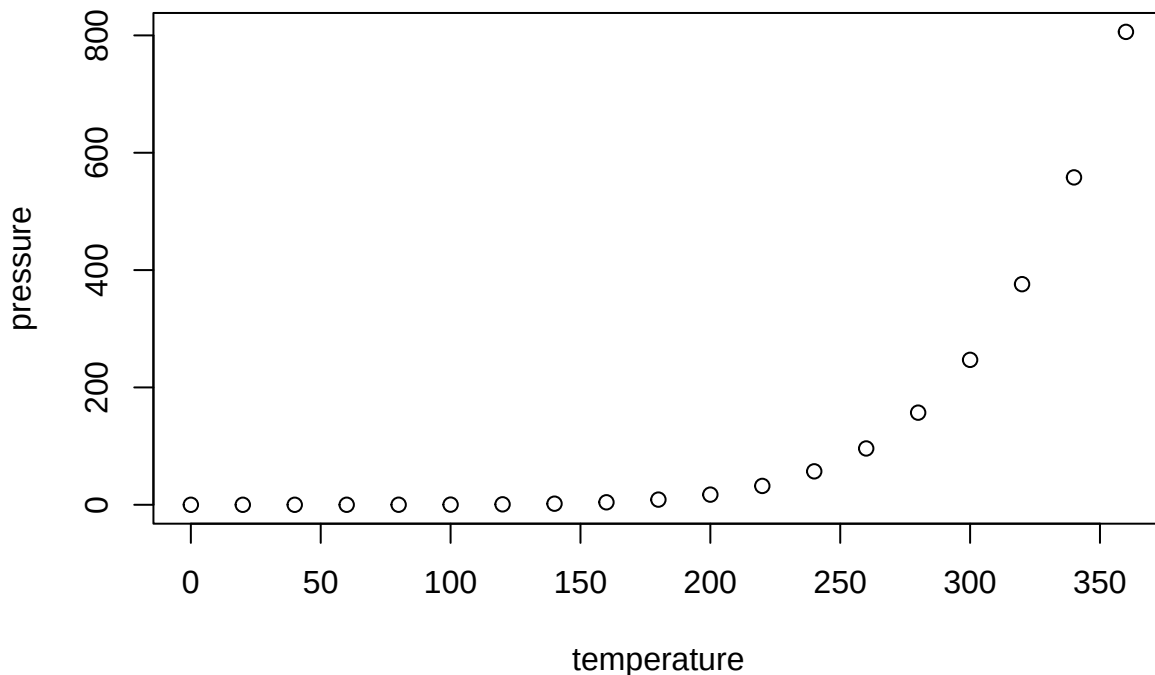
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed      dist
##  Min.   : 4.0    Min.   :  2.00
## 1st Qu.:12.0    1st Qu.: 26.00
##  Median :15.0    Median : 36.00
##   Mean  :15.4    Mean   : 42.98
## 3rd Qu.:19.0    3rd Qu.: 56.00
##   Max.  :25.0    Max.   :120.00
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.